Finite Lattice-to-$\overline{\text{MS}}$ Matching for the $SU(2)$–Higgs False-Vacuum-Decay Prefactor

Merged report: derivations, Feynman-type diagrams, project status and roadmap

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Abstract

This document merges (i) the report *"Next Steps in Finite Lattice-to-MS Matching"* (updated October 24, 2026), (ii) every Feynman-type diagram and convention figure belonging to the study, and (iii) a reader's layer added for orientation: a status dashboard, a register of open issues, a diagnostic reading of the numerical evidence, and a gated roadmap for the remaining work. Material taken from the report is presented with its original caveats; material added for this merged edition is marked **[Added]** so the two can always be told apart.

Bottom line. The lattice action, the exact vertex generators, the fixed routings and the operator projectors are complete and independently audited. Two sectors are complete for their selected scope: the scalar-only radial matching ($\delta m^2, \delta\lambda, \delta\gamma, \delta\delta, \delta c_{10}, \delta c_{12}$) and the Higgs two-point slope δZ_H. Every gauge-linked amplitude is still a pilot with QMC-only errors, the continuum gauge/ghost two-point tensor has not been computed, and the Ward/Slavnov–Taylor closure is open. The coefficient template therefore remains `null` and `includes_full_su2_lattice_to_msbar_matching = false`.

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Part I

Orientation

1 Where the project stands

1.1 [Added] Purpose in one paragraph

The project computes the one-loop prefactor of Higgs false-vacuum decay with a lattice-Euclidean determinant (4D bounce, fluctuation determinant on the lattice, $SU(2)$ gauge sector). Lattice quantities depend on the spacing a and on the lattice regularization, so parameters used in the lattice computation must be tied to renormalized $\overline{\text{MS}}$ parameters. The matching coefficients are exactly that bridge: for each local operator, the *continuum-renormalized minus lattice* one-loop amplitude evaluated at identical external momenta. Until the full set required by the chosen scope is available, the production solver cannot claim a physical prefactor; a coefficient exercise alone also cannot establish a decay-rate prefactor (collective-coordinate normalization and continuum/volume limits remain separate requirements).

1.2 [Added] Status dashboard

Sector	State	Headline number	Main caveat
Action and exact vertex generators	**complete, audited**	max discrepancy 2.4×10^{-6} (ghost table)	implemented and imported by solver, not yet consumed by matching integrator or physical determinant
Routings and projectors	**complete, audited**	tree-slope projector 1.1×10^{-15}	tests algebra only, no one-loop Ward test
Scalar-only radial matching	**complete (selected scope)**	$\delta m^2 = 2.7529$, $\delta\lambda = -1.4512$, $\delta\gamma = -2.8512$, $\delta\delta = 0.35672$, $\delta c_{10} = 1.0743$, $\delta c_{12} = 6.2049$	scalar-only, at stated a and μ_R; not full $SU(2)$
Higgs kinetic δZ_H	**complete (2-pt sector)**	$0.0329117974 \pm 3.6 \times 10^{-7}$	QMC-only error; 3-point fit is exactly determined
Scalar 4-/6-point contacts	**evaluated**	$-1.45104 \pm 5.2 \times 10^{-5}$; $-2.84684 \pm 1.8 \times 10^{-4}$	4-pt differs from zero-momentum value by $\sim 3\sigma$; 6-pt sensitive to 3-scale fit
Gauge-linked 4-Higgs	**pilot**	$\Delta\Gamma^{(4)} = 0.2540212 \pm 0.0007361$	same-sign held-out residuals $\sim 1.4\sigma$
Gauge-linked 6-Higgs	**pilot**	-2.2646 ± 0.0702	4×64 points only
Gauge-linked 8-Higgs	**paired run complete**	-19.8545 ± 1.3014 (6.56%)	finite-contact pilot; no model systematics
Gauge-linked 10-/12-Higgs	**exploratory**	$-1.238 \times 10^4 \pm 1.3 \times 10^3$; $-2.17 \times 10^6 \pm 3.6 \times 10^5$	only 4 points per scramble
Gauge two-point (continuum)	**not started**	—	required to close the Ward identity
Gauge two-point (lattice)	**open**	max normalized transverse residual 0.41; one component $23\,\sigma$	raw lattice only; IR mass breaks BRST
Coefficient JSON	**null**	—	full-matching flag stays false

Table 1: [Added] One-page status of every sector. Green: complete for the stated scope. Orange: partially evaluated. Red: exploratory or open. Gray: not available.

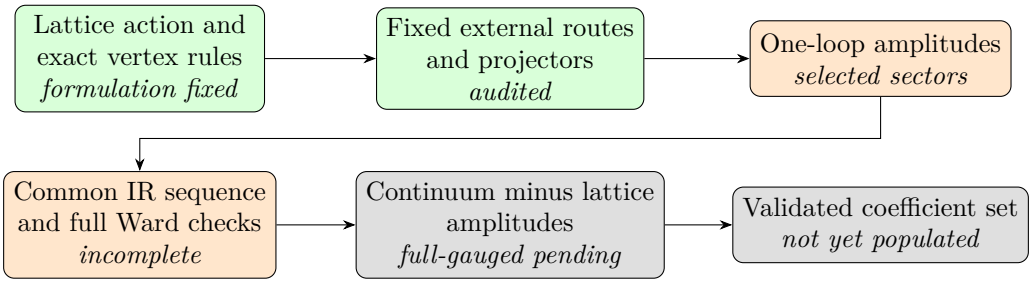


Figure 1: Status map of the finite matching program (report Fig. 1).

1.3 [Added] What is and is not established

Established.

(a) The lattice action and its quadratic expansion agree with the recorded link Hessian. (b) Every saved vertex table (scalar-link, Wilson, gauge-ghost, local scalar-ghost) matches an independent construction, and the nonzero-background Faddeev–Popov matrix matches finite gauge transformations. (c) The projector deck is nonexceptional and passes its tree-level identities. (d) The scalar-only radial coefficients are reproducible and quadrature-stable. (e) δZ_H is extracted with a clean, monotone regulator approach.

Evidence only (pilot).

All gauge-linked $2n$-point projections, and every gauge two-point diagnostic.

Not computed.

The continuum pure-gauge/ghost two-point tensor, the gauge-coupling matching, the matched zero-momentum mass coefficient, and any model-error estimate for the IR extrapolation.

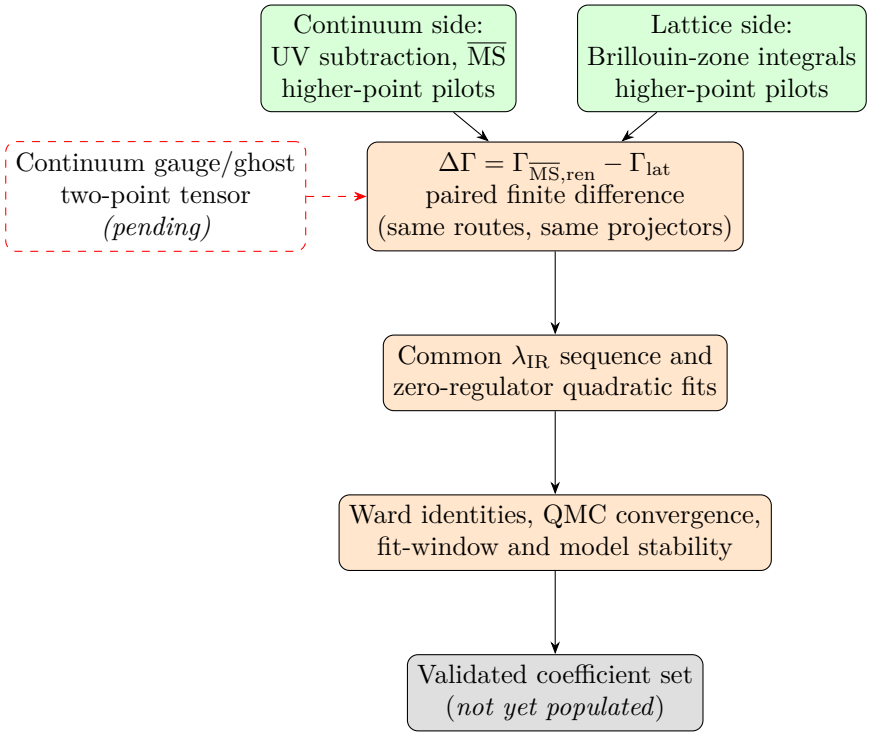


Figure 2: Finite matching requires paired continuum and lattice amplitudes at identical routes, followed by common-regulator and stability checks. Orange: partial numerical work; gray: open.

2 [Added] Reader's guide: notation, benchmark and conventions

Symbol	Meaning	Benchmark value
a	lattice spacing (solver units)	0.8
$m_0^2 = -\mu_{\rm sq}$	scalar mass squared at the false vacuum	0.4
μ_R	$\overline{\rm MS}$ matching scale, $\mu_R^2 = m_0^2$	$\sqrt{0.4} = 0.6324555\ldots$
λ, γ, δ	tree couplings of ρ^4, ρ^6, ρ^8	$-2,\ 0.8,\ 0.08$
g_0, $q = g_0/2$	bare lattice gauge coupling	0.65 (not a determination of $g_0(a)$)
ξ	background R_ξ gauge parameter	1
$\lambda_{\rm IR}$	auxiliary IR mass on massless gauge/ghost lines	$\lambda_{\rm IR}/\mu_R \in \{1/8, \ldots, 1/256\}$
$\bar\rho$	illustrative local-vertex background	0.37
$\Delta\Gamma$	$\Gamma_{\overline{\rm MS},\rm ren} - \Gamma_{\rm lat}$	sign convention throughout

Table 2: Notation and benchmark parameters. The external momentum p is a separate variable from μ_R; the default matching scale is $\mu_R = \sqrt{0.4}$, not 10.4.

[Added] Glossary. *Nonexceptional routing*: external momenta chosen so that no proper subset sums to zero, avoiding spurious IR singularities. *Local contact projector*: the intercept d_0 of a quadratic fit in the average squared momentum $\bar s$ of a contracted $2n$-point tensor. *QMC SE*: standard error across independently scrambled Sobol replicates; it measures integration noise only. *Paired*: continuum and lattice amplitudes evaluated on the same scrambled point set so that their noise is correlated and partly cancels. *Fit window*: the number of lowest regulator values (three, four or five) used in the quadratic extrapolation in $(\lambda_{\rm IR}/\mu_R)^2$. *Superficial degree D*: UV power counting of a loop cycle; $D \geq 0$ forces an $\overline{\rm MS}$ subtraction.

[Added] Reading the evidence. Throughout, three kinds of error must not be conflated: **(1)** QMC integration error (quoted); **(2)** regulator-extrapolation model error (not estimated for any gauge-linked sector); **(3)** omitted physics (gauge/ghost sectors, higher momentum dependence, gauge-coupling matching). Numbers carrying only (1) are diagnostics.

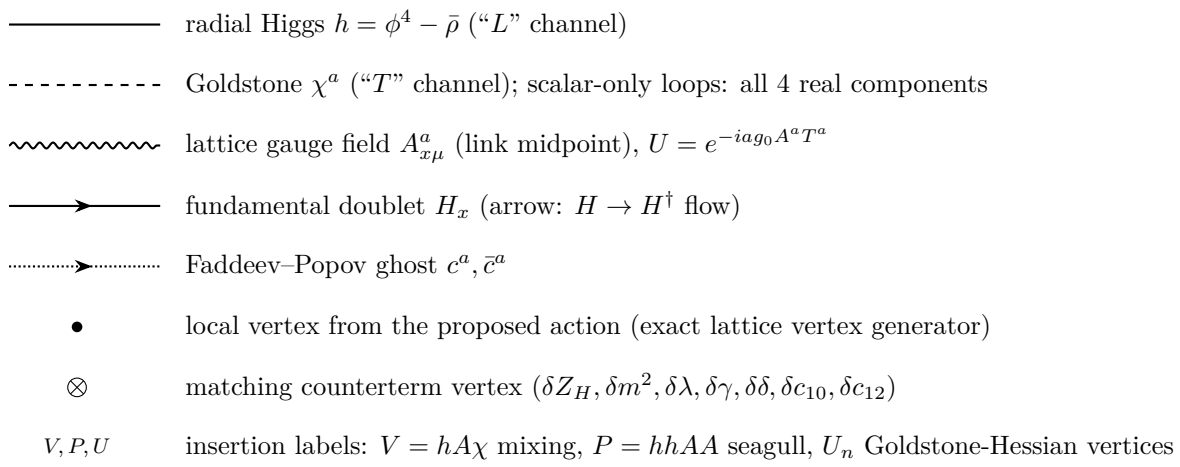


Figure 3: Line and vertex conventions used in all figures. All external momenta are incoming; site fields use $e^{+ip\cdot x}$ and gauge fields are placed at link midpoints. Lattice kernels use $\hat k_\mu = (2/a)\sin(ak_\mu/2)$.

Free propagator	Expression	Notes
Higgs/Goldstone	$\delta^{AB}/(\hat{k}^2 + m_0^2)$	$m_0^2 = -\mu_{\rm sq}$; benchmark $a = 0.8$, $m_0^2 = 0.4$
Gauge ($\xi = 1$)	$\delta_{\mu\nu}/(\hat{k}^2 + \lambda_{\rm IR}^2)$	common IR shift on both regularizations
Ghost	$1/(\hat{k}^2 + \lambda_{\rm IR}^2)$	same shift
Ghost at $\bar{\rho} \neq 0$	$\delta^{ab}/(\hat{k}^2 + \xi q^2 \bar{\rho}^2)$	$q = g_0/2$

Table 3: Free propagators attached to the lines of the figures below.

Part II

The matching study (from the report)

3 Purpose and present status (report summary)

The October 6 morning audit established that the radial counterterm basis, continuum pole residues, coefficient-input interface and prerequisite diagnostics are ready, without evaluating a finite Brillouin-zone difference. A subsequent, deliberately narrow calculation evaluated one scalar $O(4)$ quartic tadpole contribution to the mass matching; it verifies the first integral and its normalization but is not a complete mass coefficient. The report then specifies a proposed nonlinear lattice $SU(2)$ action whose quadratic expansion agrees with the recorded link Hessian. The existing determinant and fluctuation modes still use quadratic operators; an exact nonlinear vertex module is connected to the solver and generates the proposed rules.

Chronology of the calculation, as recorded:

- **Oct 8:** fixed full-gauged nonexceptional routing and projector deck generated and independently audited; first Step 2 results: Higgs two-point at the symmetric background ($\xi = 1$), scalar-only 4- and 6-point projections, scalar-matter gauge-polarization Ward check.
- **Oct 11–12:** gauge-linked four-radial-Higgs pilot (six regulators to 1/256, eight replicates of 2^{12} points); paired fit-window diagnostic with 3-/4-/5-regulator fits.
- **Oct 18:** provisional δZ_H extraction; conditional four-Higgs contact estimate; higher-point evaluators; gauge two-point follow-ups; UV pole coefficients C_6, C_8; Haar measure term.
- **Oct 19:** independent 8-leg and minimum-size 10-/12-leg pilots (QMC-only errors $\sim 62\%$, 11%, 17%).
- **Oct 23–24:** paired eight-leg run completed (eight scrambles of 2^{12} points, six regulators, three routes): three-regulator intercept -19.8545 ± 1.3014; four- and five-regulator windows -19.8636 ± 1.3033 and -19.8680 ± 1.3042.

These are finite-contact pilots whose errors exclude regulator-model systematics. No continuum pure-gauge/ghost two-point tensor or matched gauge coefficient was computed, so the full coefficient template remains null and full matching remains explicitly open.

4 Lattice action and momentum kernel

The solver uses a forward nearest-neighbor difference for the radial kinetic term,

$$S_{\rm kin}^{\rm lat} = \frac{a^4}{2} \sum_x \sum_{\nu=1}^{4} \Big(\frac{\rho_{x+a\hat{\nu}} - \rho_x}{a} \Big)^2, \tag{1}$$

acting componentwise on the four real components of the Higgs multiplet. Its free kernel is

$$\hat{k}^2 = \frac{4}{a^2}\sum_{\nu=1}^{4}\sin^2\frac{ak_\nu}{2}, \qquad k_\nu \in [-\pi/a, \pi/a], \qquad G_{\rm lat}(k) = \frac{1}{\hat{k}^2 + m^2}. \tag{2}$$

4.1 Proposed complete nonlinear lattice action

The existing fluctuation and determinant operators implement the scalar forward-difference action and the quadratic gauge–Goldstone–ghost Hessian; they do not evaluate the full nonlinear action in loop amplitudes. The solver now exposes exact nonlinear vertex generators for the following Wilson-action completion. With $T^a = \sigma^a/2$, bare coupling g_0 and a fundamental doublet H_x on each site,

$$U_{x\mu} = \exp[-iag_0 A^a_{x\mu} T^a], \tag{3}$$

$$U_{x,\mu\nu} = U_{x\mu} U_{x+a\hat{\mu},\nu} U^\dagger_{x+a\hat{\nu},\mu} U^\dagger_{x\nu}, \tag{4}$$

on the perturbative branch $A^a_{x\mu} = 2i\,\mathrm{Tr}[T^a \log U_{x\mu}]/(ag_0)$. The gauge action is the Wilson plaquette action [1],

$$S_g = \frac{4}{g_0^2}\sum_x \sum_{\mu<\nu}\Big(1 - \tfrac{1}{2}\mathrm{Re\,Tr}\, U_{x,\mu\nu}\Big), \tag{5}$$

whose continuum normalization is $\frac{1}{4}\int d^4x\, F^a_{\mu\nu}F^a_{\mu\nu}$.

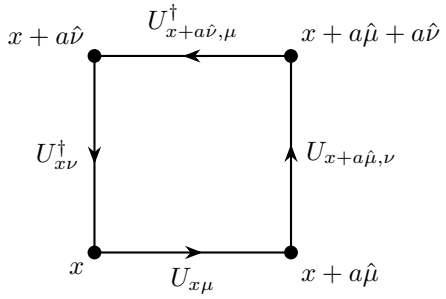


Figure 4: Oriented Wilson plaquette $U_{x,\mu\nu}$ with action weight $\frac{4}{g_0^2}\big(1-\frac{1}{2}\mathrm{Re\,Tr}\, U_{x,\mu\nu}\big)$. Its ordered four-edge expansion generates all pure-gauge vertices.

With $\rho_x = \sqrt{2H_x^\dagger H_x}$, the covariant scalar action is

$$S_{H,\rm link} = a^4 \sum_{x,\mu} \frac{(U_{x\mu}H_{x+a\hat{\mu}} - H_x)^\dagger (U_{x\mu}H_{x+a\hat{\mu}} - H_x)}{a^2}. \tag{6}$$

At $U = 1$, $H_x = (\phi^2_x + i\phi^1_x,\ \phi^4_x - i\phi^3_x)^T/\sqrt{2}$ reproduces the canonical $\frac{1}{2}a^4\sum_{x,\mu,A}[(\phi^A_{x+a\hat{\mu}} - \phi^A_x)/a]^2$. The tree potential and one-loop-order counterterms are

$$V_0(\rho) = -\tfrac{1}{2}\mu_{\rm sq}\rho^2 + \tfrac{1}{4}\lambda\rho^4 + \tfrac{1}{6}\gamma\rho^6 + \tfrac{1}{8}\delta\rho^8, \quad m_0^2 = -\mu_{\rm sq}, \tag{7}$$

$$V_{\rm ct}(\rho) = \tfrac{1}{2}\delta m^2\rho^2 + \tfrac{1}{4}\delta\lambda\rho^4 + \tfrac{1}{6}\delta\gamma\rho^6 + \tfrac{1}{8}\delta\delta\rho^8 + \tfrac{1}{10}\delta c_{10}\rho^{10} + \tfrac{1}{12}\delta c_{12}\rho^{12}, \tag{8}$$

with the gauge-covariant kinetic counterterm $\delta Z_H\, S_{H,\rm link}$. The dimension-10 and -12 operators are perturbative matching counterterms, not leading tree interactions.

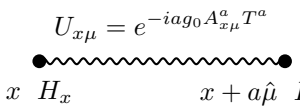


Figure 5: Scalar-link term $S_{H,x\mu} = \frac{a}{2}\big[H_+^\dagger H_+ + H_x^\dagger H_x - H_+^\dagger e^{+iag_0 A} H_x - H_x^\dagger e^{-iag_0 A} H_+\big]$. Each scalar–scalar–$r$-gauge vertex is the r-th derivative of the link exponentials at $A = 0$.

Gauge fixing and ghosts. Around a radial background $\bar{\rho}_x$ with vanishing background gauge field use the background R_ξ gauge $F_x^a = (DA^a)_x - \xi q \bar{\rho}_x \chi_x^a$, $S_{\rm gf} = \frac{a^4}{2\xi}\sum (F_x^a)^2$, $q = g_0/2$, with D the backward divergence. The Faddeev–Popov operator is the exact variation of F under $H_x \mapsto G_x H_x$, $U_{x\mu} \mapsto G_x U_{x\mu} G^\dagger_{x+a\hat{\mu}}$, $G_x = \exp[-ig_0 \omega_x^a T^a]$, and $S_{\rm gh} = a^4 \sum \bar{c}_x^a M^{ab}_{xy} c_y^b$ with $M^{ab}_{xy} = \delta F_x^a / \delta \omega_y^b$. The complete perturbative action is

$$S = S_g + S_{H,\rm link} + a^4 \sum_x [V_0(\rho_x) + V_{\rm ct}(\rho_x)] + S_{\rm gf} + S_{\rm gh}. \tag{9}$$

The gauge-fixing and ghost terms define the perturbative background-field calculation and are not part of the gauge-invariant bare action. The coupling stays g_0 in the link variables until bare-to-renormalized matching is specified.

4.2 Quadratic check against the implemented Hessian

Expanding about $H_x = (0, \bar{\rho}_x)^T/\sqrt{2}$ and $H_x = (\chi_x, \bar{\rho}_x + h_x)^T/\sqrt{2}$ in a single real generator plane,

$$S^{(2)}_{H,\rm link} = \frac{a^4}{2} \sum_{x,\mu} \Big[\frac{(\chi_{x+a\hat{\mu}} - \chi_x)^2}{a^2} + q^2 \bar{\rho}_x \bar{\rho}_{x+a\hat{\mu}} A^2_{x\mu} \Big] \tag{10}$$

$$+ a^4 \sum_{x,\mu} \frac{q}{a} A_{x\mu} (\bar{\rho}_{x+a\hat{\mu}} \chi_x - \bar{\rho}_x \chi_{x+a\hat{\mu}}), \tag{11}$$

which agrees with the link-Hessian expression in the validation report. At $\bar{\rho} = 0$ the Higgs and three Goldstones have mass m_0, gauge fields and ghosts are massless, and with $\hat{k}_\mu = 2\sin(ak_\mu/2)/a$ the free kernels are

$$\begin{aligned} G^{AB} &= \frac{\delta^{AB}}{\hat{k}^2 + m_0^2}, \qquad K_{\mu\nu} = \hat{k}^2 \delta_{\mu\nu} - (1 - \xi^{-1}) \hat{k}_\mu \hat{k}_\nu, \\ D_{\mu\nu} &= \frac{1}{\hat{k}^2} \Big[\delta_{\mu\nu} - (1 - \xi) \frac{\hat{k}_\mu \hat{k}_\nu}{\hat{k}^2} \Big], \qquad G_{\rm gh} = \frac{1}{\hat{k}^2}. \end{aligned} \tag{12}$$

For $\xi = 1$, $D_{\mu\nu} = \delta_{\mu\nu}/\hat{k}^2$. The zero-momentum massless modes need a declared IR and gauge-volume prescription.

5 Step 1: exact vertex generators

The expressions below define the lattice vertices without replacing the link action by a continuum derivative expansion. They are implemented in `su2_lattice_vertices.py` and imported by the solver; they generate the displayed one-loop vertex tables but are not yet consumed by the matching integrator or the physical determinant.

5.1 Scalar–gauge vertices

For one oriented link with $H_+ = H_{x+a\hat{\mu}}$ and $A = A^a_{x\mu} T^a$,

$$S_{H,x\mu} = \frac{a}{2}\Big[H_+^\dagger H_+ + H_x^\dagger H_x - H_+^\dagger e^{+iag_0 A} H_x - H_x^\dagger e^{-iag_0 A} H_+\Big], \tag{13}$$

so every scalar–scalar–r-gauge vertex is the r-th derivative of the two exponentials at $A = 0$; e.g.

$$S^{(A)}_{H,x\mu} = ia^3 g_0 (H_x^\dagger A H_+ - H_+^\dagger A H_x), \qquad S^{(A^2)}_{H,x\mu} = \frac{a^4 g_0^2}{2}(H_+^\dagger A^2 H_x + H_x^\dagger A^2 H_+). \tag{14}$$

Use $\phi_x = \int_{\mathrm{BZ}} \frac{d^4p}{(2\pi)^4} e^{ip\cdot x}\phi(p)$ for site fields and the same with $x \to x + a\hat{\mu}/2$ for $A_{x\mu}$; all momenta are incoming and conserved modulo a reciprocal lattice vector. With incoming labels p' on $H^\dagger$, p on H, gauge legs k_i, define $\theta_\mu = \frac{a}{2}(p' - p)_\mu$ and $T^{a_1\cdots a_r} = \frac{1}{r!}\sum_\pi T^{a_{\pi(1)}} \cdots T^{a_{\pi(r)}}$. Then, for all gauge legs polarized along μ,

$$V^{a_1\cdots a_r}_{H^\dagger H A^r_\mu} = -\frac{(iag_0)^r}{a^2}\big[e^{i\theta_\mu} + (-1)^r e^{-i\theta_\mu}\big] T^{a_1\cdots a_r}, \tag{15}$$

and the first rules are $V_{H^\dagger H} = \hat{p}^2$, $V^a_{H^\dagger H A^a_\mu} = \frac{2g_0}{a}\sin\theta_\mu T^a$, $V^{ab}_{H^\dagger H A^a_\mu A^b_\mu} = \frac{g_0^2}{2}\cos\theta_\mu\, \delta^{ab}$⊮.

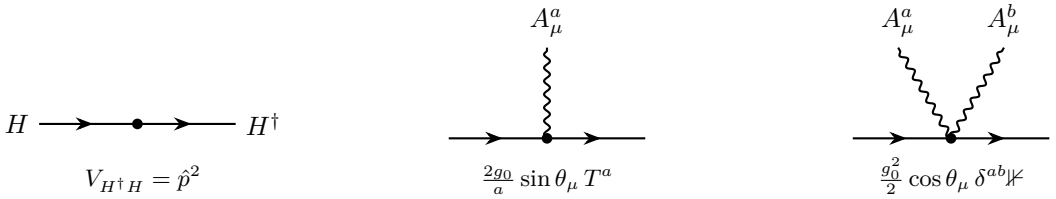


Figure 6: Scalar-link vertices with endpoint phases $\theta_\mu = \frac{a}{2}(p' - p)_\mu$; general r: $-\frac{(iag_0)^r}{a^2}[e^{i\theta_\mu} + (-1)^r e^{-i\theta_\mu}]T^{a_1\cdots a_r}$, nonzero only for gauge legs polarized along the link direction.

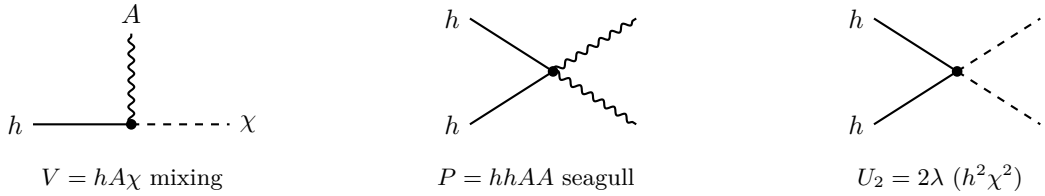


Figure 7: Insertions entering the coupled gauge–scalar Hessian $K_B[h]$ in a radial source $h(x)$. Higher Goldstone-Hessian vertices $U_4 = 4!\,\gamma\ (h^4\chi^2)$ and $U_6 = 6!\,\delta\ (h^6\chi^2)$ have the same topology with four and six h legs.

5.2 Pure-gauge vertices

All pure-gauge vertices follow from differentiating the plaquette weight. Label the four ordered edges of a $\mu\nu$ plaquette $j = 1, \ldots, 4$ with axes $(\alpha_j) = (\mu, \nu, \mu, \nu)$, orientation signs $(\epsilon_j) = (+, +, -, -)$ and midpoint offsets $(d_j) = (\frac{a\hat{\mu}}{2},\ a\hat{\mu} + \frac{a\hat{\nu}}{2},\ a\hat{\nu} + \frac{a\hat{\mu}}{2},\ \frac{a\hat{\nu}}{2})$. For n labeled legs (a_i, μ_i, k_i) assigned to edges σ_i with $\alpha_{\sigma_i} = \mu_i$, set $I_j = \{i : \sigma_i = j\}$, $m_j = |I_j|$ and

$$B_j(I_j) = \frac{(-iag_0\epsilon_j)^{m_j}}{m_j!}\sum_{\pi\in S_{m_j}} T^{a_{\pi(1)}}\cdots T^{a_{\pi(m_j)}}, \qquad B_j(\emptyset) = 1, \tag{16}$$

$$V^{a_1\cdots a_n}_{\mu_1\cdots\mu_n}(k_1,\dots,k_n) = -\frac{2}{a^4 g_0^2}\sum_{\mu<\nu}\sum_{\sigma_i}\Big(\prod_i e^{ik_i\cdot d_{\sigma_i}}\Big)\mathrm{Re}\,\mathrm{Tr}[B_1B_2B_3B_4]. \tag{17}$$

Color-algebra checks use $\mathrm{tr}(T^aT^b)=\frac{1}{2}\delta^{ab}$, $\mathrm{tr}(T^aT^bT^c)=\frac{i}{4}\epsilon^{abc}$, $\mathrm{tr}(T^aT^bT^cT^d)=\frac{1}{8}(\delta^{ab}\delta^{cd}-\delta^{ac}\delta^{bd}+\delta^{ad}\delta^{bc})$. The quadratic vertex summed over plaquettes gives the transverse kernel; adding gauge fixing gives $K_{\mu\nu}$.

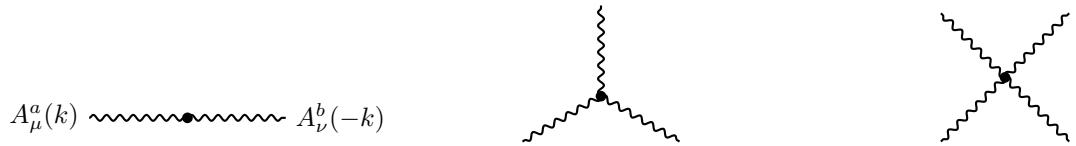


Figure 8: Wilson-plaquette n-gauge vertices generated by the ordered-edge formula $V^{a_1\cdots a_n}_{\mu_1\cdots\mu_n}(k_1,\dots,k_n)$ with exact edge phases $e^{ik_i\cdot d_{\sigma_i}}$; $n\ge 5$ follow from the same generator.

5.3 Ghost vertices

In the logarithmic link coordinate, $t^a=-iT^a$, $X=ag_0A^at^a=\log U$, $C=\mathrm{ad}X$ and under $U_{x\mu}\mapsto G_xU_{x\mu}G^\dagger_{x+a\hat\mu}$, $G_x=\exp(g_0\omega^a_xt^a)$ the exact variation is $\delta X=\frac{C}{e^C-1}[g_0\omega_x-e^Cg_0\omega_{x+a\hat\mu}]$, i.e.

$$\begin{aligned}\delta A_{x\mu}=\frac{1}{a}\Big[&(\omega_x-\omega_{x+a\hat\mu})-\frac{ag_0}{2}A_{x\mu}\times(\omega_x+\omega_{x+a\hat\mu})\\&+\frac{(ag_0)^2}{12}A\times\big(A\times(\omega_x-\omega_{x+a\hat\mu})\big)-\frac{(ag_0)^4}{720}A^{\times4}(\omega_x-\omega_{x+a\hat\mu})+\cdots\Big].\end{aligned} \tag{18}$$

With $H=(\phi^2+i\phi^1,\phi^4-i\phi^3)^T/\sqrt{2}$, $\chi^a=\phi^a$, $h=\phi^4-\bar\rho$: $\delta\chi^a=q[-(\bar\rho+h)\delta^{ab}+\epsilon^{abc}\chi^c]\omega^b$, $\delta h=q\chi^b\omega^b$. Then $M^{ab}_{xy}=\sum_\mu D_{x\mu}\delta A^a_{x\mu}/\delta\omega^b_y-\xi q\bar\rho_x\delta\chi^a_x/\delta\omega^b_y$, and for constant $\bar\rho$ and vanishing fluctuations $M^{ab}_0(k)=\delta^{ab}(\hat k^2+\xi q^2\bar\rho^2)$. The scalar ghost vertices are $\bar c^ac^bh:+\xi q^2\bar\rho\,\delta^{ab}$ and $\bar c^ac^b\chi^c:-\xi q^2\bar\rho\,\epsilon^{abc}$; the one-gauge vertex is $V_{\bar c^ac^bA^c_\mu}=-ig_0\epsilon^{abc}\hat p_\mu\cos\frac{ak_\mu}{2}$. Using a linearized continuum transformation instead would change these vertices.

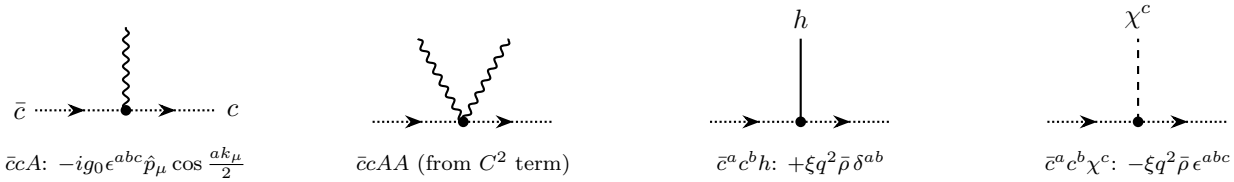


Figure 9: Faddeev–Popov vertices from the exact variation of the logarithmic link coordinate, $\delta X=\frac{C}{e^C-1}[g_0\omega_x-e^Cg_0\omega_{x+a\hat\mu}]$. Higher $\bar ccA^r$ vertices follow from the full series of $C/(e^C-1)$.

5.4 Scalar-potential vertices

With $X=\phi^A\phi^A$ and $f(X)=-\frac{1}{2}\mu_{\rm sq}X+\frac{1}{4}\lambda X^2+\frac{1}{6}\gamma X^3+\frac{1}{8}\delta X^4$, all vertices follow from $V_{A_1\cdots A_n}=\partial_{A_1}\cdots\partial_{A_n}f$:

$$V_{A_1\cdots A_n}=\sum_{j=0}^{\lfloor n/2\rfloor}2^{n-j}f^{(n-j)}(X)\sum_{P_j}\Big(\prod_{\{r,s\}\in P_j}\delta_{A_rA_s}\Big)\Big(\prod_{t\ \text{unpaired}}\phi^{A_t}\Big), \tag{19}$$

including the radial and transverse curvatures $U_L=m_0^2+3\lambda\rho^2+5\gamma\rho^4+7\delta\rho^6$ and $U_T=\rho^{-1}V_0'=m_0^2+\lambda\rho^2+\gamma\rho^4+\delta\rho^6$. At vanishing background the nonzero tree vertices are $V^{(2)}=m_0^2\delta$, $V^{(4)}=2\lambda\mathcal{P}_2$,

$V^{(6)} = 8\gamma\mathcal{P}_3$, $V^{(8)} = 48\delta\mathcal{P}_4$, with $\mathcal{P}_r$ the sum over pairings; counterterm vertices have the same tensor structure with coefficients $\delta m^2, 2\delta\lambda, 8\delta\gamma, 48\delta\delta, 384\delta c_{10}, 3840\delta c_{12}$.



Figure 10: Momentum-independent scalar tree vertices at $\rho = 0$ ($\mathcal{P}_r$: sum over pairings of the $O(4)$ indices). Vertices at nonzero background follow from $\partial_{A_1}\cdots\partial_{A_n} f(X)$, giving the curvatures $U_L = m_0^2 + 3\lambda\rho^2 + 5\gamma\rho^4 + 7\delta\rho^6$ and $U_T = m_0^2 + \lambda\rho^2 + \gamma\rho^4 + \delta\rho^6$.

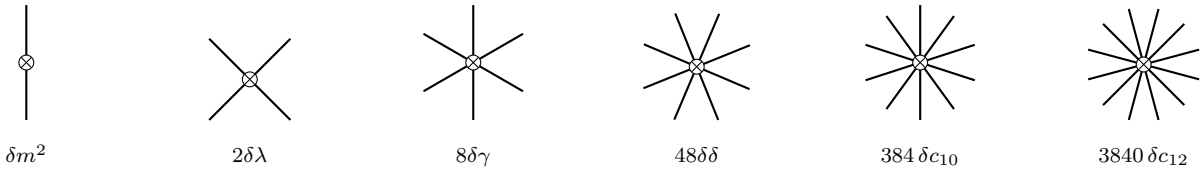
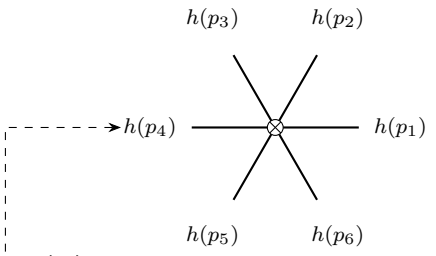


Figure 11: Counterterm vertices, one-loop order (not inserted in one-loop propagators or loops), with $2, 4, \ldots, 12$ scalar legs. The kinetic counterterm $\delta Z_H\, S_{H,\text{link}}$ is momentum dependent and not drawn.

(A) The vertex: six external legs, nothing attached to them

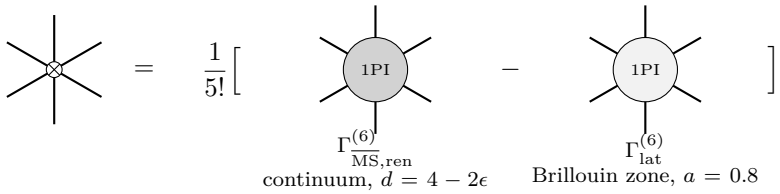


Counterterm for the ρ^6 coefficient: vertex factor $8\,\delta\gamma\,\mathcal{P}_3$, with $\mathcal{P}_3$ the sum over the 15 pairings of the six $O(4)$ indices.

Legs are *amputated external* scalar lines, incoming momenta, $\sum_i p_i = 0$, at the zero-sum nonexceptional shapes $\bar{s} \in \{0.2, 0.4, 0.6\}$.

The vertex is of one-loop order, so it is not inserted into any loop or propagator; it must equal $\Delta\Gamma^{(6)}/5!$.

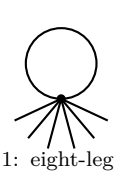
(B) What the vertex must equal



Both blobs: one loop, same six momenta, same $\lambda_{\rm IR}$ sequence; continuum-renormalized *minus* lattice.

(C) What is inside each blob (representative diagrams; all labeled-leg partitions are summed)

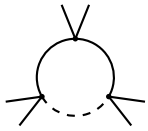
Scalar-only loops (radial solid, Goldstone dashed); $\delta\gamma = 6\Delta v_3$ split by ring order j [Added]



$j = 1$: eight-leg δ vertex, one loop closed; weight $10\delta\,\Delta I_1$; contributes -0.55059



$j = 2$: λ vertex (2 legs) and γ vertex (4 legs); weight $36\lambda\gamma\,\Delta I_2$; contributes -2.67348



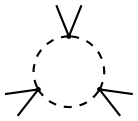
$j = 3$: three λ vertices, two legs each; weight $30\lambda^3\,\Delta I_3$ (finite); contributes $+0.37287$

Sum $= -2.85119512 = \delta\gamma^{\rm scalar}$ (zero-momentum projector); $\Delta I_n = I_n^{\overline{\rm MS}} - I_n^{\rm lat}$.

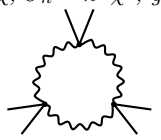
Gauge-linked loops (insertion classes $P = hhAA$, $V = hA\chi$, $U_n = h^n\chi^2$; gauge lines wavy)



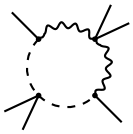
$U_6 = 6!\,\delta$ Goldstone tadpole



U_2^3, $U_2 = 2\lambda$ Goldstone triangle

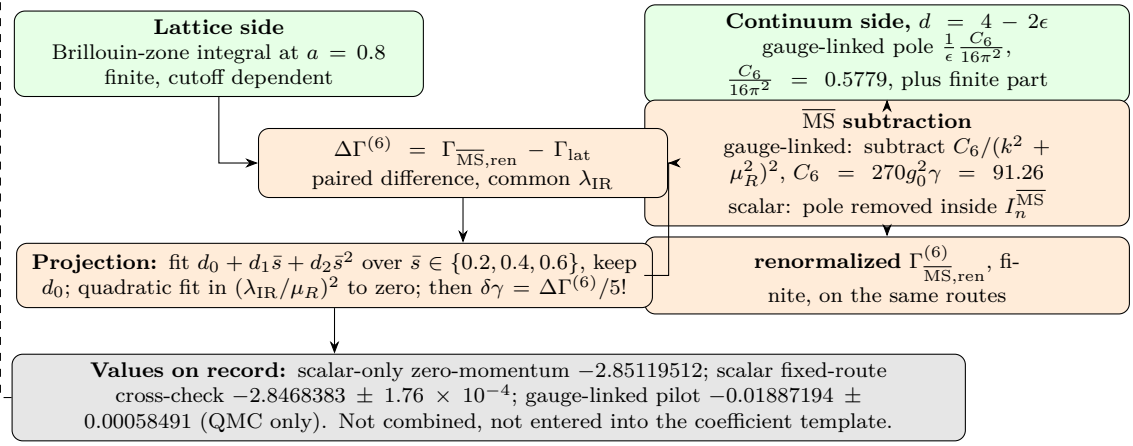


P^3 gauge triangle (seagulls)



$P\,V\,U_2\,V$ mixed gauge–Goldstone ring

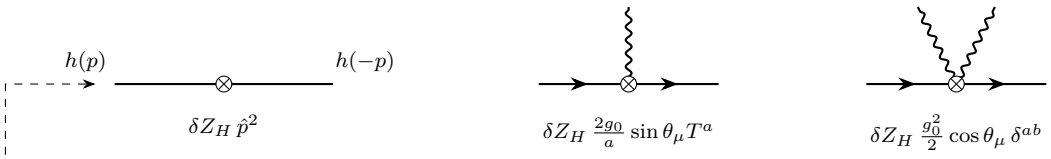
(D) Renormalization and matching path for this one coefficient



coefficient $8\delta\gamma$ enters panel (A) once validated

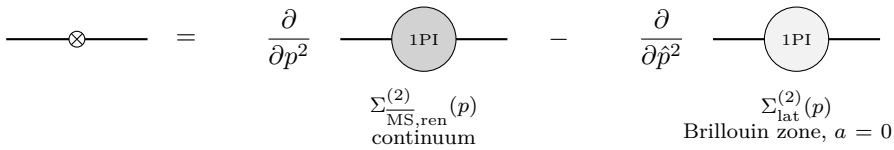
Figure 12: [Added] Anatomy of the six-leg counterterm vertex ($\delta\gamma$, ρ^6 coefficient) of Fig. 11: (A) vertex; (B) identity it implements; (C) one-loop families inside the two blobs; (D) renormalization and matching path. Topologies are representative; the split by ring order is arithmetic on the tabulated moments and closes to the report value.

(A) The vertex: two legs, momentum dependent, with gauge-covariant companions



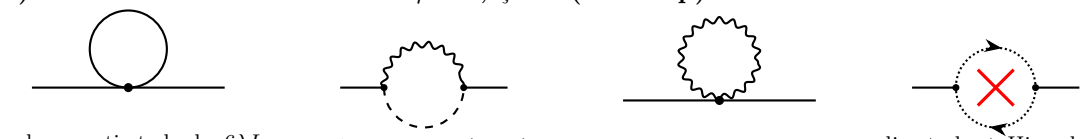
The counterterm is $\delta Z_H\, S_{H,\mathrm{link}}$, so it multiplies the whole gauge-covariant scalar-link term: the two-point piece $\hat{p}^2$ and the scalar–scalar–gauge pieces share the same coefficient. Unlike $\delta\gamma$, it depends on external momentum.

(B) What the vertex must equal: a slope difference, not an intercept



$\delta Z_H(\mu) = \partial_{p^2}\Sigma_{\overline{\mathrm{MS}},\mathrm{ren}}\big|_{p^2=\mu^2} - \partial_{\hat{p}^2}\Sigma_{\mathrm{lat}}\big|_{\hat{p}^2=\mu^2}$, at $\mu^2 = \mu_R^2 = 0.4$; lattice coordinate $\hat{p}^2 = 0.3966567139724254$. A first derivative, with no $1/n!$.

(C) What is inside each blob at $\bar{\rho} = 0$, $\xi = 1$ (one loop)



scalar quartic tadpole, $6\lambda I$
momentum independent
slope $= 0$, $\delta Z_H^{\mathrm{scalar}} = 0$

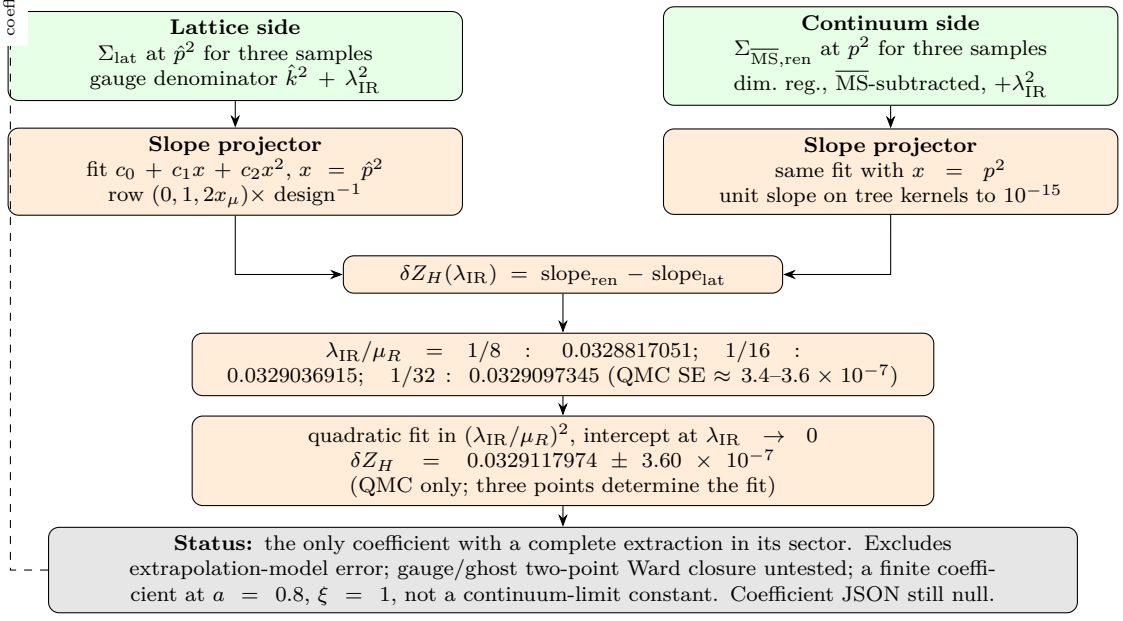
$hA\chi$ exchange (V–V)
gauge line massless: $+\lambda_{\mathrm{IR}}^2$
Goldstone line massive

$hhAA$ seagull (P)
gauge loop, $+\lambda_{\mathrm{IR}}^2$

direct ghost–Higgs loop
vanishes at $\bar{\rho} = 0$
in this gauge

Same external momenta $p = \sqrt{s}\,\hat{n}$, $s \in \{0.2, 0.4, 0.6\}$ on both sides. The slope is carried by the gauge-linked diagrams, hence the common IR regulator.

(D) Renormalization and matching path



coefficient enters the vertices of panel (A)

Figure 13: [Added] Anatomy of the kinetic counterterm δZ_H for contrast with the $\delta\gamma$ page: (A) the momentum-dependent two-leg vertex and its gauge-covariant companions; (B) the slope identity; (C) the one-loop families at $\bar{\rho} = 0$; (D) the path to the extracted value. Topologies are representative.

	$\delta\gamma$ (six legs)	δZ_H (two legs)
Vertex	$8\,\delta\gamma\,\mathcal{P}_3$, momentum independent	$\delta Z_H\,\hat{p}^2$; multiplies all of $S_{H,\mathrm{link}}$ (covariant companions)
Identity	$\delta\gamma = \Delta\Gamma^{(6)}/5!$	$\delta Z_H = \partial_{p^2}\Sigma_{\overline{\mathrm{MS}},\mathrm{ren}} - \partial_{\hat{p}^2}\Sigma_{\mathrm{lat}}$
Projector	intercept d_0 of a fit in $\bar{s}$ at zero-sum six-leg routes	slope from a fit in p^2 (or $\hat{p}^2$) at $p^2 = \mu_R^2$
Loop content	scalar rings ($j = 1, 2, 3$) plus gauge-linked insertion classes	gauge exchange and seagull only; scalar tadpole has zero slope; ghost–Higgs vanishes
IR regulator	needed only for the gauge-linked part	needed, since the whole slope is gauge-linked
UV treatment	scalar: inside $I_n^{\overline{\mathrm{MS}}}$; gauge-linked: subtract $C_6/(k^2+\mu_R^2)^2$	$\overline{\mathrm{MS}}$-subtracted continuum self-energy
Value on record	-2.85119512 (scalar, zero momentum); gauge-linked pilot $-0.01887194 \pm 0.00058491$	$0.0329117974 \pm 3.6 \times 10^{-7}$
Status	scalar part complete, gauge-linked part pilot	complete for the two-point sector (QMC only)

Table 4: [Added] Side-by-side contrast of the two counterterm anatomies. Values are those on record in the report; none is combined or entered into the coefficient template.

5.5 Local validation of the vertex module

Check (at $a = 0.8$, $g_0 = 0.65$, $\xi = 1$)	Max. abs. discrepancy
Momentum conservation, all generated tables	0
Scalar-link Taylor coefficients (orders 1–2)	8.020×10^{-9}
Complete scalar-link momentum/color table	6.204×10^{-9}
Wilson-plaquette Taylor coefficients (orders 2–4)	1.719×10^{-9}
Complete Wilson tensor vs. independent expansion	0
Wilson leg-permutation symmetry	4.442×10^{-16}
Wilson quadratic kernel vs. transverse Hessian	7.216×10^{-16}
Log-link gauge variation	5.954×10^{-11}
Real-Higgs transformation	1.813×10^{-10}
Scalar-link / plaquette gauge-invariance Ward checks	1.776×10^{-15} / 2.220×10^{-16}
Complete gauge-ghost table vs. finite transformations	2.398×10^{-6}
Local scalar-ghost tensors	1.735×10^{-18}
Nonzero-background FP matrix (absolute / relative)	3.234×10^{-10} / 1.679×10^{-11}
Free Faddeev–Popov kernel	7.105×10^{-15}

Table 5: Step 1 vertex audit. The saved tables cover every index component at their listed routings; arbitrary-momentum functions remain available. These are implementation checks, not perturbative uncertainty estimates, and do not sample every external momentum.

5.6 October 8 fixed full-gauged routing and projector deck

Generated by `su2_full_gauged_projection_setup.py` (setup $a = 0.8$, $g_0 = 0.65$, $\xi = 1$, $\mu_R^2 = 0.4$, $\bar{\rho} = 0.37$). Two-point legs $(p, -p)$ use $p = \sqrt{s}\,\hat{n}$, $\hat{n} = (1, 2, 3, 4)/\sqrt{30}$, $s \in \{0.2, 0.4, 0.6\}$; at $p^2 = \mu_R^2 = 0.4$ the lattice coordinate is $\hat{p}^2 = 0.3966567139724254$. The asymmetric scalar–scalar–gauge triangle uses $p_{H^\dagger} = \sqrt{0.4}(1, 0, 0, 0)$, $p_H = \sqrt{0.4}(0.2, 0.4, 0.8, 0)$, $p_A = -p_{H^\dagger} - p_H$, with squared norms $0.4, 0.336, 0.896$. Local $2n$-scalar contacts use fixed-shape zero-sum routings at $\bar{s} \in \{0.2, 0.4, 0.6\}$ with every proper subset sum nonzero.

Projectors. For the Higgs two-point function fit $c_0 + c_1 x + c_2 x^2$ with $x = p^2$ (continuum) or $\hat{p}^2$ (lattice); the slope projector is the row $(0, 1, 2x_\mu)$ times the inverse design matrix (unit slope on both tree kernels to 1.11×10^{-15}). For gauge two-point tensors, with $K = p$ (continuum) or $K_\mu = 2\sin(ap_\mu/2)/a$ (lattice), $T_{\mu\nu} = K^2\delta_{\mu\nu} - K_\mu K_\nu$, $L_{\mu\nu} = K_\mu K_\nu$,

$$P_T^{\mu\nu} = \frac{T^{\mu\nu}}{3(K^2)^2}, \qquad P_L^{\mu\nu} = \frac{K^\mu K^\nu}{(K^2)^2}. \tag{20}$$

For each local $2n$-scalar contact the complete tensor is contracted first, then fit to $d_0 + d_1\bar{s} + d_2\bar{s}^2$ retaining d_0 (separate continuum and lattice design matrices). *Common IR prescription:* background Feynman gauge on both regularizations with $k^2 \mapsto k^2 + \lambda_{\mathrm{IR}}^2$ and $\hat{k}^2 \mapsto \hat{k}^2 + \lambda_{\mathrm{IR}}^2$ on every massless gauge and ghost denominator; massive scalar denominators are unshifted.

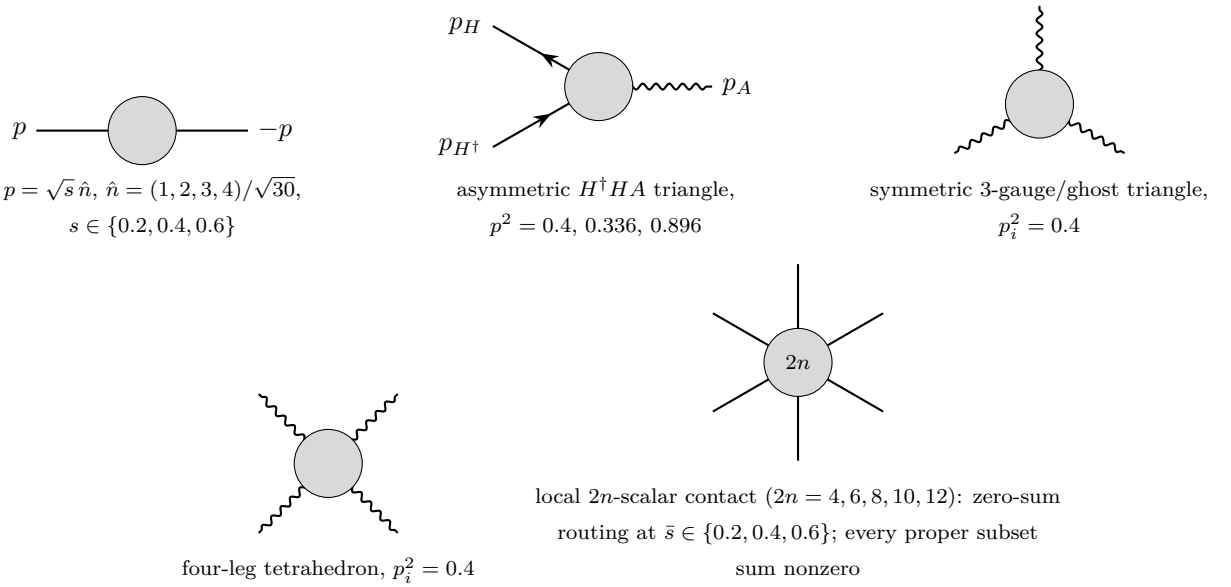


Figure 14: Fixed nonexceptional routings for which the lattice and continuum projections are evaluated. The two-point slope is projected with $(0, 1, 2x_\mu)\cdot$(inverse design matrix); gauge two-point tensors are contracted with $P_T^{\mu\nu} = T^{\mu\nu}/[3(K^2)^2]$, $P_L^{\mu\nu} = K^\mu K^\nu/(K^2)^2$; contact tensors are contracted first and fit to $d_0 + d_1\bar{s} + d_2\bar{s}^2$, retaining d_0.

Audit item (`su2_full_gauged_projection_audit.py`: PASS)	Max. discrepancy
Momentum conservation / smallest proper-subset norm	4.65×10^{-16} / 7.14×10^{-2}
Scalar-link tensors vs. independent finite-action differences	2.90×10^{-9}
Asymmetric scalar–scalar–gauge Ward identity	3.82×10^{-17}
Wilson tensors vs. ordered-word expansion	1.33×10^{-15}
Wilson two-point transversality / gauge projectors	6.11×10^{-16} / 2.19×10^{-15}
Ghost tensors vs. exact finite transformations	2.72×10^{-6}
Nonzero-background FP matrix (abs. / rel.)	3.23×10^{-10} / 1.68×10^{-11}
Higgs tree-slope / local-contact projector identities	1.11×10^{-15} / 8.88×10^{-16}

Table 6: Independent audit of the routing and projector deck. These do not amount to an exhaustive momentum sweep, a one-loop Ward-identity test, or full-gauged matching.

6 Step 2: one-loop projections

For a constant radial background ρ, zero background gauge field and $\xi = 1$, set $M_A^2 = q^2\rho^2$. Up to a ρ-independent constant the lattice one-loop potential is

$$V_{1,\mathrm{lat}}(\rho) = \frac{1}{2}\int_{\mathrm{BZ}} \frac{d^4k}{(2\pi)^4}\Big[\log(\hat{k}^2 + U_L) + 3\log(\hat{k}^2 + U_T + M_A^2) + 6\log(\hat{k}^2 + M_A^2)\Big], \tag{21}$$

with the radial mode, three Goldstones, and the net gauge-plus-ghost contribution. The continuum comparison is

$$V_{1,\mathrm{cont}}(\rho) = \frac{1}{2}\Big(\frac{e^{\gamma_E}\mu_R^2}{4\pi}\Big)^\epsilon \int \frac{d^dk}{(2\pi)^d}\Big[\log(k^2 + U_L) + 3\log(k^2 + U_T + M_A^2) + 3(d-2)\log(k^2 + M_A^2)\Big], \tag{22}$$

retaining the d-dependent vector/ghost multiplicity through $\overline{\text{MS}}$ subtraction; setting it to its four-dimensional value before subtracting can lose a finite term. The determinant construction follows the background-field framework [3], and the nonanalytic gauge contribution is the Coleman–Weinberg one [4].

Kinetic projection.

$$\delta Z_H(\mu) = \frac{\partial}{\partial p^2}\Sigma^{(2)}_{\overline{\text{MS}},\text{ren}}(p)\Big|_{p^2=\mu^2} - \frac{\partial}{\partial \hat{p}^2}\Sigma^{(2)}_{\text{lat}}(p)\Big|_{\hat{p}^2=\mu^2}. \tag{23}$$

Mass and higher potential coefficients are projections of $\Gamma^{(2n)}_{\overline{\text{MS}},\text{ren}} - \Gamma^{(2n)}_{\text{lat}}$ onto local $O(4)$-invariant operators at identical external momenta, with a projector separating potential from momentum-dependent operators. The constant-background determinant is a useful check, but its Taylor expansion at $\rho = 0$ is *not* a valid full-gauge-theory potential projector: massless gauge modes generate nonanalytic $\rho^4 \log \rho^2$ terms. No full-gauged finite coefficient is inferred from the determinant formula or from partial checks.

6.1 Evaluated sector: Higgs two-point function

At $\bar{\rho} = 0$ the one-loop Higgs two-point diagrams are the scalar quartic tadpole and gauge exchange plus seagull; the Faddeev–Popov operator has no Higgs coupling at this background in the selected gauge. With $a = 0.8$, $g_0 = 0.65$, $m_0^2 = \mu_R^2 = 0.4$, $\lambda = -2$, $\xi = 1$ and scrambled Sobol integration [7, 8] on the lattice, dimensional regularization with $\overline{\text{MS}}$ [5, 6] in the continuum:

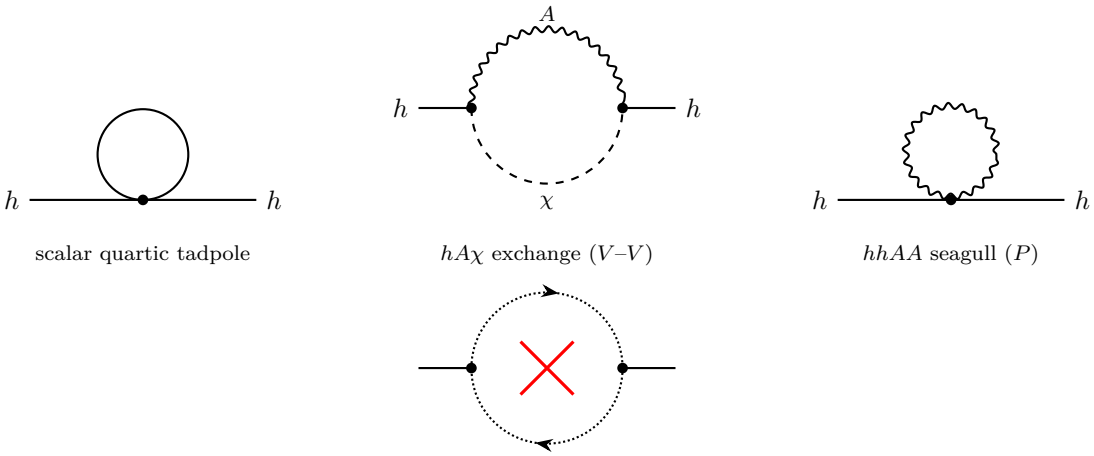


Figure 15: Diagram families for $\Sigma^{(2)}$ of the Higgs field at the symmetric background in $\xi = 1$. The result $\delta Z_H = 0.0329117974 \pm 3.6 \times 10^{-7}$ (QMC only, quadratic fit in $(\lambda_{\text{IR}}/\mu_R)^2$) comes from gauge exchange plus seagull; the scalar tadpole has no momentum dependence.

$\lambda_{\rm IR}/\mu_R$	$\delta Z_H^{\overline{\rm MS}-{\rm lat}}$	QMC SE
1/8	0.0328817051	3.41×10^{-7}
1/16	0.0329036915	3.45×10^{-7}
1/32	0.0329097345	3.55×10^{-7}
0 (quadratic fit in $(\lambda_{\rm IR}/\mu_R)^2$)	0.0329117974	3.60×10^{-7}

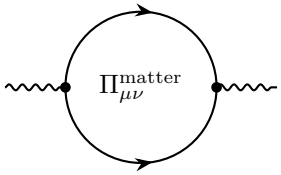
Table 7: Higgs slope finite difference. Three points determine the quadratic exactly, so there is no independent fourth-point model check; the monotone approach is evidence of regulator stability for this combination, not proof of cancellation in the uncomputed gauge/ghost sectors.

6.2 Evaluated sector: scalar-only contacts and the Ward check

Amplitude (fixed nonexceptional routes)	Projected $\overline{\rm MS}-$lattice	QMC error
Four scalar legs	-1.4510393	5.16×10^{-5}
Six scalar legs	-2.8468383	1.76×10^{-4}

Table 8: Scalar-only finite-momentum cross-checks; they include all four real scalar components but no gauge-linked diagrams. The six-point value is sensitive to omitted higher momentum dependence in the three-scale quadratic fit and must not be substituted for a full-gauged or exact zero-momentum coefficient.

At the symmetric point $F^a = \partial \cdot A^a$ has an FP operator independent of the Higgs background, so no direct ghost loop enters Higgs-only amplitudes at $\bar{\rho} = 0$ (ghost loops remain necessary in gauge-field amplitudes). The scalar-matter polarization at $p^2 = 0.4$ has a continuum normalized transverse residual 1.15×10^{-15}; on the lattice it is 4.77×10^{-4} with the four contracted components within 1.06 QMC SE of zero. This is a matter-loop check only.



$p_\mu \, \Pi_{\mu\nu}^{\rm matter}(p) \simeq 0$
normalized residual 1.15×10^{-15} (continuum), 4.8×10^{-4} (lattice)

Figure 16: Scalar-matter polarization Ward check at $p^2 = 0.4$; a matter-loop check only, not a Slavnov–Taylor test.

6.3 Organizing the remaining Higgs-only amplitudes

In a general radial source $h(x) = \phi^4(x)$,

$$\Gamma_B^{(1)}[h] = \tfrac{1}{2}{\rm Tr}\log K_B[h], \qquad K_B[h] = \begin{pmatrix} K_A^{(0)} + \Pi_{AA}[h^2] & V_{A\phi}[h] \\ V_{\phi A}[h] & K_\phi^{(0)} + U''(h\hat{e}_4) \end{pmatrix}, \tag{24}$$

with twelve gauge and four real scalar components, $U_T'' = m_0^2 + \lambda h^2 + \gamma h^4 + \delta h^6$, $U_L'' = m_0^2 + 3\lambda h^2 + 5\gamma h^4 + 7\delta h^6$. The $2n$-point integrands are the $2n$-th source derivatives at $h = 0$, evaluated as the finite

insertion expansion

$$\Gamma_B^{(1)}[h] - \Gamma_B^{(1)}[0] = \frac{1}{2}\sum_{j\geq 1} \frac{(-1)^{j+1}}{j} \mathrm{Tr}\big[(G_0 \Delta K_B[h])^j\big], \quad G_0 = K_B[0]^{-1}. \tag{25}$$

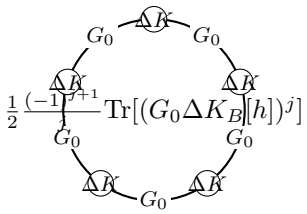


Figure 17: Generic bosonic ring. $K_B[h] = \begin{pmatrix} K_A^{(0)} + \Pi_{AA}[h^2] & V_{A\phi}[h] \\ V_{\phi A}[h] & K_\phi^{(0)} + U''(h) \end{pmatrix}$ with 12 gauge and 4 real scalar components. The $2n$-point integrands are the $2n$-th source derivatives of $\Gamma_B^{(1)}[h] = \frac{1}{2}\mathrm{Tr}\log K_B[h]$ at $h = 0$; each ΔK insertion is one of V, P or U_n.

6.4 Pilot: gauge-linked four-Higgs projection

The insertion sum has the PP, PVV, UVV and VVVV terms ($V = hA\chi$, $P = hhAA$, $U = h^2\chi^2$); direct Higgs–ghost diagrams vanish at the symmetric background. The continuum integrand is $\overline{\mathrm{MS}}$-subtracted by $C/(k^2+\mu_R^2)^2$, $C = -54q^4 + 36q^2\lambda = -8.2074586$ (zero renormalized integral at μ_R).

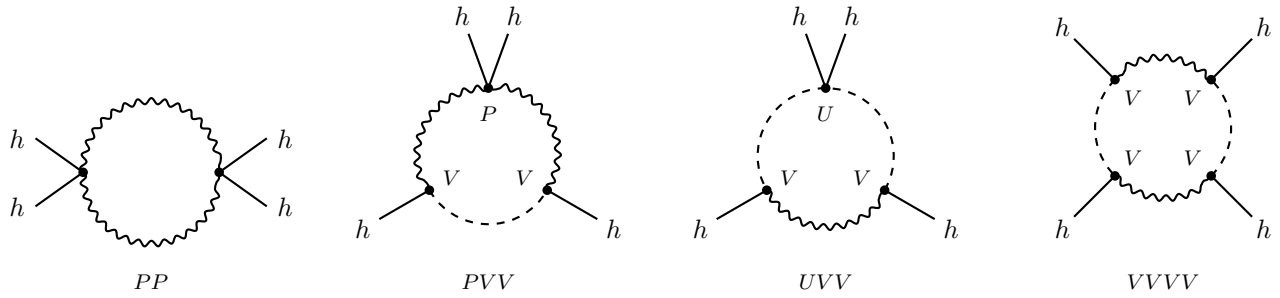


Figure 18: The four insertion families of the gauge-linked four-Higgs pilot ($P = hhAA$, $V = hA\chi$, $U = h^2\chi^2$). Direct Higgs–ghost diagrams vanish at the symmetric background. Continuum integrand is $\overline{\mathrm{MS}}$-subtracted by $C/(k^2+\mu_R^2)^2$, $C = -54q^4+36q^2\lambda = -8.2074586$. Pilot: $\delta\lambda_{\mathrm{gauge}} = \Delta\Gamma^{(4)}/3! \simeq +0.0423$ (QMC only; not adopted).

$\lambda_{\rm IR}/\mu_R$	$\Delta\Gamma^{(4)}_{\rm gauge}$	QMC SE
1/8	0.2542605	3.34×10^{-4}
1/16	0.2541718	5.43×10^{-4}
1/32	0.2540645	6.81×10^{-4}
1/64 (held out)	0.2539542	7.53×10^{-4}
1/128 (held out)	0.2538960	7.83×10^{-4}
1/256 (held out)	0.2538767	7.92×10^{-4}

Table 9: October 11 reproduction, eight replicates of 2^{12} points per route and regulator. The quadratic fit to the first three values gives intercept 0.2540212 ± 0.0007361 (QMC only). Held-out residuals (observed minus predicted) are $-7.82\times10^{-5}\pm5.45\times10^{-5}$, $-1.28\times10^{-4}\pm9.03\times10^{-5}$, $-1.45\times10^{-4}\pm1.03\times10^{-4}$, each ~ 1.4 SE below prediction. With $\delta\lambda = \Delta\Gamma^{(4)}/3!$ the gauge-sector contact diagnostic is $+0.04234\pm0.00012$ (not entered into any coefficient file).

Fit-window diagnostic (Oct 12, paired, lower statistics). Three-, four- and five-regulator quadratic fits give intercepts 0.2541409 ± 0.0015671, 0.2539852 ± 0.0016464, 0.2538930 ± 0.0016921. The paired shifts are -0.0001557 ± 0.0001113 (four vs. three) and -0.0002479 ± 0.0001753 (five vs. three), both ~ 1.4 paired SE below zero, unresolved at this precision. Held-out residuals of the three-point fit at $1/64, 1/128, 1/256$ are -0.0002190 ± 0.0001565, -0.0003620 ± 0.0002543, -0.0004110 ± 0.0002872. The same-sign pattern is a fit-window warning, not a bound on extrapolation-model error.

Integrity checks. The insertion sum reproduces the constant-source gauge/Goldstone determinant to below 7×10^{-13}; the nonzero-momentum $hA\chi$ and $hhAA$ rules agree with the exact link generator to below 6×10^{-17}; the large-momentum integrand reproduces C/k^4 to relative error 8.1×10^{-7}; the projected integrand is Bose symmetric. All errors are QMC-only replicate errors.

7 First evaluated diagram: scalar quartic tadpole

Take $V_4(\Phi) = \frac{\lambda}{4}(\Phi^A\Phi^A)^2$, $A = 1, \ldots, 4$. For an $O(N)$ multiplet the zero-momentum tadpole self-energy is $(N+2)\lambda I$, so $\Sigma_{\rm tad} = 6\lambda I$ for $N = 4$. With $q_\nu = ak_\nu$, $x = a^2m^2$,

$$I_{\rm lat}(a,m) = \int_{[-\pi/a,\pi/a]^4} \frac{d^4k}{(2\pi)^4} \frac{1}{\hat{k}^2 + m^2} = \frac{1}{a^2} J(x), \quad J(x) = \int_0^\infty dt\, e^{-xt} \big[e^{-2t} I_0(2t)\big]^4, \tag{26}$$

and the renormalized continuum tadpole is $I_{\overline{\rm MS}}(m,\mu) = \frac{m^2}{16\pi^2}[\ln(m^2/\mu^2) - 1]$. Equating zero-momentum two-point functions, $m^2_{\rm lat} + \Sigma_{\rm lat}(0) = m^2_{\overline{\rm MS}} + \Sigma^{\rm ren}_{\overline{\rm MS}}(0)$ and $m^2_{\rm lat} = m^2_{\overline{\rm MS}} + \delta m^2_{\rm tad}$, giving for this diagram only $\delta m^2_{\rm tad} = 6\lambda(I_{\overline{\rm MS}} - I_{\rm lat})$.

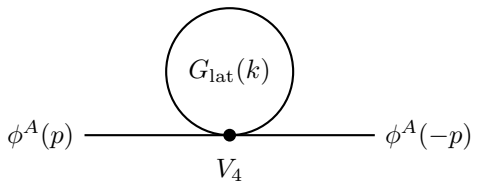


Figure 19: Scalar quartic tadpole: $\Sigma_{\rm tad} = (N+2)\lambda I = 6\lambda I$, giving $\delta m^2_{\rm tad} = 6\lambda(I_{\overline{\rm MS}} - I_{\rm lat}) = 2.7529$ at the benchmark. This diagram alone does not determine δm^2 or δZ_H (momentum independent, $\delta Z_H^{\rm scalar} = 0$).

Quantity ($a = 0.8$, $m^2 = \mu^2 = 0.4$, $\lambda = -2$)	Value
$J(a^2m^2)$	0.1452018039938607
I_{lat}	0.2268778187404072
$I_{\overline{\text{MS}}}$	−0.002533029591058444
δm^2_{tad}	2.752930179977589

Table 10: Quartic-tadpole check. Adaptive quadrature error on J is 2.45×10^{-14} ($\sim 4.6\times10^{-13}$ in δm^2_{tad}); this excludes perturbative truncation. The large additive shift is a cutoff-dependent mass contribution, not a universal finite constant.

What this example does not determine. It must not be entered as the solver `delta_m2`: it is one diagram. It has no momentum dependence and cannot fix δZ_H; nor does it determine the $\rho^4, \ldots, \rho^{12}$ coefficients, gauge-field normalization, gauge-coupling matching, or a decay-rate prefactor.

8 Completed Steps 1–2: scalar-only radial matching

This section selects a finite, reproducible scope: retain the $O(4)$ tree potential through ρ^8, compute its one-loop scalar determinant, project the local potential basis through ρ^{12}, with gauge, link and ghost loops excluded. All scalar propagators have mass m_0 at $\rho = 0$, so zero external momenta give a valid local projector. With $z = \rho^2$, $D = \hat{k}^2$,

$$P_L(z) = 3\lambda z + 5\gamma z^2 + 7\delta z^3, \qquad P_T(z) = \lambda z + \gamma z^2 + \delta z^3, \tag{27}$$

$$V^{\text{scalar}}_{1,\text{lat}}(z) = \frac{1}{2}\int_{\text{BZ}} \frac{d^4k}{(2\pi)^4}\big[\log(D + m_0^2 + P_L) + 3\log(D + m_0^2 + P_T)\big]. \tag{28}$$

Define $I^{\text{lat}}_n = \int_{\text{BZ}}(\hat{k}^2 + m_0^2)^{-n} = \frac{a^{2n-4}}{(n-1)!}\int_0^\infty dt\, t^{n-1}e^{-a^2m_0^2t}[e^{-2t}I_0(2t)]^4$ and the continuum moments $I_1^{\overline{\text{MS}}} = \frac{m_0^2}{16\pi^2}[\log\frac{m_0^2}{\mu_R^2} - 1]$, $I_2^{\overline{\text{MS}}} = -\frac{1}{16\pi^2}\log\frac{m_0^2}{\mu_R^2}$, $I_n^{\overline{\text{MS}}} = \frac{\Gamma(n-2)}{16\pi^2\Gamma(n)}(m_0^2)^{2-n}$ for $n \geq 3$. Then

$$\Delta v_n = \frac{1}{2}\sum_{j=1}^{n}\frac{(-1)^{j+1}}{j}(I_j^{\overline{\text{MS}}} - I_j^{\text{lat}})[z^n](P_L^j + 3P_T^j), \qquad \delta c_{2n} = 2n\,\Delta v_n, \tag{29}$$

where δc_{2n} for $n = 1, \ldots, 6$ denotes $\delta m^2, \delta\lambda, \delta\gamma, \delta\delta, \delta c_{10}, \delta c_{12}$. Equivalently $\delta c_{2n} = \Delta\Gamma^{(2n)}_{1\cdots1}/(2n-1)!$ at zero momentum. The sign is continuum-renormalized minus lattice.

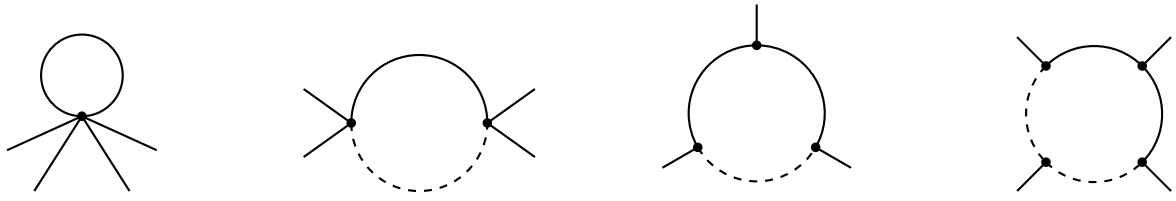


Figure 20: One-loop scalar-only ring diagrams generating $\delta c_{2n} = 2n\,\Delta v_n$, $\Delta v_n = \frac{1}{2}\sum_{j=1}^{n}\frac{(-1)^{j+1}}{j}(I_j^{\overline{\text{MS}}} - I_j^{\text{lat}})[z^n](P_L^j + 3P_T^j)$. Each internal line is a radial (L) or one of three Goldstone (T) propagators with mass m_0; the external legs carry zero momentum in the local projector. Examples: $\delta m^2 = 6\lambda\Delta I_1$ ($j = 1$), $\delta\lambda_{\text{scalar},0} = 16\gamma\,\Delta I_1 - 12\lambda^2\Delta I_2$ ($j = 1, 2$ with γ and λ vertices, drawn above).

n	I_n^{lat}	$I_n^{\overline{\text{MS}}}$	Coefficient	Value
1	0.2268778187404072	−0.002533029591058444	δm^2	2.75293017998
2	0.03094303011979438	0	$\delta\lambda$	−1.45119341289
3	0.009469337125601956	0.007915717472057639	$\delta\gamma$	−2.85119512150
4	0.007100564330706442	0.006596431226714698	$\delta\delta$	0.356724233481
5	0.008539339433655424	0.008245539033393373	δc_{10}	1.07431690553
6	0.01264685637877819	0.01236830855009006	δc_{12}	6.20491819587

Table 11: Benchmark $a = 0.8$, $m_0^2 = \mu_R^2 = 0.4$, $\lambda = -2$, $\gamma = 0.8$, $\delta = 0.08$. The mass entry reproduces the independent tadpole. Re-running adaptive quadrature at tolerances $2 \times 10^{-12}/2 \times 10^{-13}$ changed each coefficient by $< 1.1 \times 10^{-15}$ (a quadrature-stability check, not a truncation uncertainty). At this order the scalar-only two-point graph is a momentum-independent tadpole, so $\delta Z_H^{\text{scalar}} = 0$.

These are radial scalar-only results at the stated a and scale, not universal constants and not full-$SU(2)$ coefficients; they must not enter the full-matching coefficient template. The gauge determinant cannot simply be Taylor-expanded at $\rho = 0$ to extend this table.

9 Coefficient-level extraction and higher-point pilots (Oct 18–24)

9.1 October 18: initial coefficient-level extraction

Using $\Delta\Gamma = \Gamma_{\overline{\text{MS}},\text{ren}} - \Gamma_{\text{lat}}$ at the benchmark: $\delta Z_H^{\text{1-loop}}(a = 0.8, \mu_R = \sqrt{0.4}, \xi = 1) = 0.0329117974$, $\sigma_{\text{QMC}} = 3.60 \times 10^{-7}$ (a finite coefficient for the specified slope projector, not a continuum-limit constant). Normalization checks on the scalar sector give $\delta m^2_{\text{tad}} = 6\lambda\Delta I_1 = 2.75293018$ and $\delta\lambda_{\text{scalar},0} = 16\gamma\Delta I_1 - 12\lambda^2\Delta I_2 = -1.45119341$.

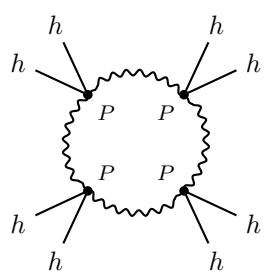
At the fixed nonexceptional routes $\delta\lambda_{\text{scalar,contact}} = -1.45103929$ (SE 5.16×10^{-5}) and the gauge-linked pilot gives $\delta\lambda_{\text{gauge,pilot}} = +0.04233687$ (SE 1.23×10^{-4}, via $\delta\lambda = \Delta\Gamma^{(4)}/6$). The exploratory sum is

$$\delta\lambda_{\text{contact,pilot}} = -1.40870242, \quad \sigma_{\text{QMC}} = 1.33 \times 10^{-4}, \tag{30}$$

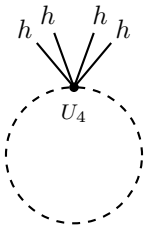
assuming independent sector randomizations and excluding all systematics. The nonexceptional scalar contact differs from the zero-momentum scalar result by 1.54×10^{-4}, about three quoted standard errors; this unresolved closure check makes the sum a preliminary diagnostic, not a coefficient. Only δZ_H has a complete extraction in its sector.

9.2 October 18: higher-point implementation pilot

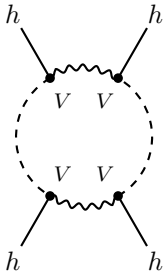
The evaluator `su2_higher_point_gauge_insertions_20261018.py` supports the fixed 4-, 6-, 8-, 10-, 12-Higgs routes, including the Goldstone-Hessian vertices $U_2 = 2\lambda$, $U_4 = 4!\gamma$, $U_6 = 6!\delta$ and omitting the pure scalar-only Goldstone determinant from the gauge-linked sector. The generalized trace-log closes against the audited four-point sum at zero source to below 6.8×10^{-13}. A raw-lattice six-leg pilot (two replicates of four Sobol points) gave regulator-to-regulator movement (0.542 ± 0.002, -3.13 ± 2.94, -1.07 ± 2.32) too large to support an IR extrapolation; the 8/10/12-leg results at that stage were single-loop-momentum smoke tests.



pure-P ring: gauge loop with seagull insertions



U_2, U_4, U_6 Goldstone loop with radial-potential insertion



alternating gauge–Goldstone loop of paired V insertions

Figure 21: Representative classes of the trace-log insertion sum used by the higher-point evaluator (Fig. 8 of the report), for $L = 6, 8, 10, 12$ external radial legs. All allowed labeled-leg partitions, cycle orderings and symmetry factors are summed numerically. For a cycle with m propagators and v derivative $hA\chi$ vertices the superficial degree is $D = 4 - 2m + v$; at $L = 10, 12$ all cycles have $D \leq -2$, so the $\overline{\text{MS}}$ pole coefficients $C_{10} = C_{12} = 0$, while $C_6 = 270 g_0^2 \gamma$ and $C_8 = 15120 g_0^2 \delta$.

9.3 October 18: UV subtraction and six-Higgs matched pilot

An angular high-momentum audit at $R = 256, 512, 1024$ with linear $1/R^2$ extrapolation gives $C_6 = 91.2599992$ and $C_8 = 511.056028$, with coupling-scaling closed forms $C_6 = 270 g_0^2 \gamma$, $C_8 = 15120 g_0^2 \delta$. The continuum $1/\epsilon$ poles are $C_6/16\pi^2 = 0.5779107$ and $C_8/16\pi^2 = 3.2363001$ ($\overline{\text{MS}}$ counterterms carry the opposite sign). The integrand subtracts $C_n/(k^2 + \mu_R^2)^2$, whose finite integral vanishes at μ_R, and is then differenced against the lattice BZ result following [9].

Six-Higgs matched pilot (four scrambles, 64 points per replicate and route, six common regulators): three-, four- and five-point-fit intercepts -2.2646325 ± 0.0701896, -2.2644884 ± 0.0699967, -2.2644328 ± 0.0699178; paired shifts 0.000144 ± 0.000265 and 0.000200 ± 0.000376 (unresolved). With $\delta\gamma_{\text{gauge}} = \Delta\Gamma^{(6)}/5!$: $\delta\gamma_{\text{gauge}} = -0.01887194 \pm 0.00058491$ (QMC only; excludes scalar-only loops and all extrapolation and UV-asymptotic systematics; keep separate from the scalar-only six-point result).

9.4 Ten- and twelve-leg UV degree

For a gauge-linked cycle with m propagators and v derivative $hA\chi$ vertices the continuum superficial degree is $D = 4 - 2m + v$. Enumerating the allowed P, U_2, U_4, U_6 and paired-V cycles at ten and twelve legs gives $D \leq -2$ (pure-P cycles are more convergent), so there is no logarithmic $\overline{\text{MS}}$ pole and $C_{10} = C_{12} = 0$ by one-loop lattice power counting [10]. This removes only the pole-subtraction term; the finite integrals, projections and IR limits are still required.

9.5 October 19: fixed-route higher-point finite-contact pilots

Higgs legs	QMC design (scrambles × points)	Three-point intercept	Relative QMC SE
8	8×32	-23.0962 ± 14.4191	62.4%
10	8×4	$-1.23780 \times 10^4 \pm 1.31017 \times 10^3$	10.6%
12	8×4	$-2.16743 \times 10^6 \pm 3.61715 \times 10^5$	16.7%

Table 12: October 19 diagnostics (three-regulator window; replicate errors only). For $L = 8$ the 4-/5-regulator windows shift the intercept by $+0.00172 \pm 0.00411$ and $+0.00236 \pm 0.00577$; relative to the earlier 4×16 run (-63.9587 ± 43.2906) it moves by $+40.86 \pm 45.63$ (~ 0.90 combined SE). For $L = 10$ the five-vs-three shift is $+1.187 \pm 1.215$; for $L = 12$, $+121.5 \pm 113.8$. The earlier 2×4 pilots had much smaller replicate errors (144 and 2.91×10^5), showing those errors should not be read as stable precision.

9.6 October 18–19: gauge two-point and link-measure follow-up

For logarithmic link coordinates the $SU(2)$ Haar density is $J(A) = [\sin(ag_0|A|/2)/(ag_0|A|/2)]^2$, contributing $-\log J = (ag_0)^2 A^a A^a/12 + O(A^4)$ and $\Delta\Pi^{\mathrm{Haar}}_{\mu\nu} = \frac{g_0^2}{6a^2}\delta_{\mu\nu}$, i.e. 0.110026 at $g_0 = 0.65$, $a = 0.8$. Including it in the 4-scramble, 32-point gauge two-point follow-up leaves a maximum normalized residual 0.4669 (one component 4.37 SE). In the seed-nested follow-up (four scrambles, 128 points) the maximum projected QMC SE falls from 0.1714 to 0.0534 but the maximum normalized transverse residual remains 0.4123; at $p^2 = 0.4$, $\lambda_{\mathrm{IR}}/\mu_R = 1/32$ one Ward component is -0.01198 ± 0.00052 (23.1 SE). The auxiliary IR mass breaks BRST and no continuum gauge tensor is subtracted, so this is a raw-lattice diagnostic, neither a Ward pass nor a continuum failure. Direct-versus-cached cubic-tensor contraction agrees to 8.9×10^{-16}. The earlier 2×4 pilot (max normalized transverse residual 0.57) was too noisy to support any conclusion.

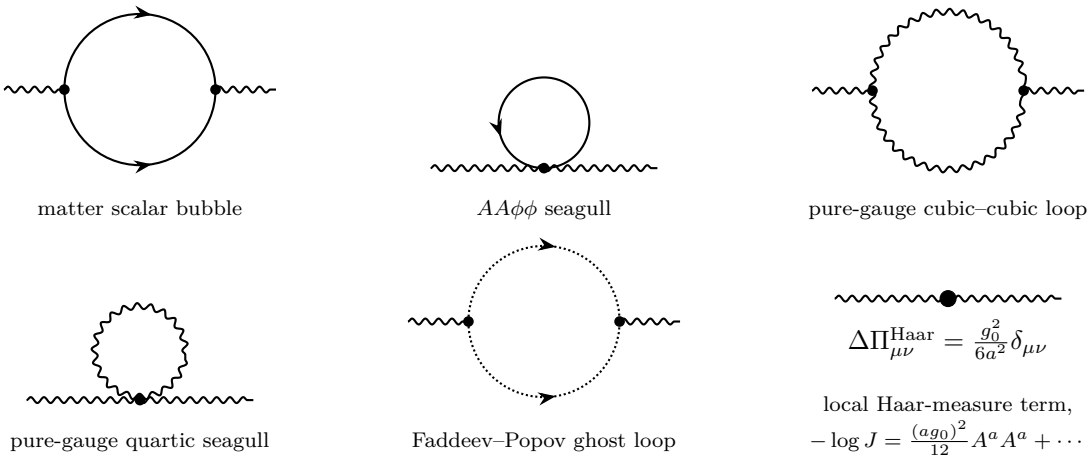


Figure 22: Bookkeeping for the one-loop gauge two-point function in logarithmic link coordinates. The Haar box is a local term, not a propagating loop. The matter-only Ward check (scalar bubble plus seagull; massive lines only, no IR regulator) passes; the pure-gauge, ghost and Haar pieces are so far raw-lattice diagnostics only, and the continuum gauge/ghost tensor is missing.

9.7 October 23–24: paired eight-leg QMC follow-up

The restartable paired driver completed the eight-Higgs fixed-route projection with eight scrambled-Sobol replicates of $4096 = 2^{12}$ points at each of six regulators and three routes; continuum and

lattice evaluations share each scrambled point set, common across routes and regulators. Run ID `20261023T173421Z`: all $144 = 8 \times 6 \times 3$ tasks and all 576 checkpoint blocks complete (2026-10-24 01:25 UTC), reaggregation reproduces saved arrays to roundoff, wall time 28271.3 s ($\approx$ 7.85 h), result SHA-256 `c909808d3b53fd531e54742f26dd007a8d35bc3cb48ee32d7c5d32dcb07c289d`. With $C_8 = 15120 g_0^2 \delta = 511.056$, $C_8/16\pi^2 = 3.23630$:

Fit window	Intercept	Replicate QMC SE	Paired shift from 3-point fit
3 regulators	-19.85454	1.30141	0
4 regulators	-19.86357	1.30335	-0.00903 ± 0.00373
5 regulators	-19.86800	1.30421	-0.01347 ± 0.00560

Table 13: Paired eight-leg result; three-regulator relative SE 6.56%. The window shifts are small relative to the intercept but nonzero on their paired-QMC scale (about 2.4σ for the five-regulator shift); neither they nor the replicate SE estimate regulator-model systematics. This is a finite-contact projection, not an accepted matching coefficient.

Part III

Assessment and roadmap

10 [Added] Diagnostic reading of the numerical evidence

This section applies the reviewer-style reading requested for the project: it flags weak points, assumptions and gaps. It uses only numbers already stated in the report.

Cross-run consistency of the eight-leg sector. The three independent eight-leg determinations are -63.96 ± 43.29 (4×16), -23.10 ± 14.42 (8×32) and -19.85 ± 1.30 (8×2^{12}). The last two differ by 3.24 against a combined error of 14.5 ($\approx 0.2\sigma$), and the first differs from the last by $\approx 1.0\sigma$. The sector is therefore statistically self-consistent, and the precision gain from 2^5 to 2^{12} points per replicate ($14.42 \to 1.30$, a factor ≈ 11) is encouraging for the QMC convergence rate. The same improvement has not been demonstrated for $L = 10, 12$.

Regulator-extrapolation model. Every gauge-linked sector uses a quadratic in $x = (\lambda_{\mathrm{IR}}/\mu_R)^2$. Three observations bear on this: (i) in the four-point sector the three held-out points all lie $\sim 1.4\sigma$ below the prediction, a coherent rather than random pattern; (ii) the $L = 8$ paired window shifts are $\sim 2.4\sigma$ at five regulators; (iii) with massless lines shifted by λ_{IR}^2, the amplitude dependence on λ_{IR} may include non-analytic terms (e.g. $x \log x$) that a pure polynomial in x would absorb into its coefficients. The report does not test this. Since the three-point quadratic is exactly determined, it carries no internal check at all.

Closure of the scalar sector. The nonexceptional scalar four-point contact ($-1.45103929 \pm 5.16 \times 10^{-5}$) differs from the zero-momentum scalar result (-1.45119341) by 1.54×10^{-4}, $\approx 3\sigma$. Because both are scalar-only with massive lines, regulator effects are absent; the likely candidates are residual momentum dependence in the three-scale quadratic projector or QMC under-coverage. This is the cleanest available test of the projector itself, and it must close before the gauge-linked contact values are added to scalar ones.

Gauge Ward closure. The raw-lattice gauge two-point residual is not a failure or pass: the IR mass breaks BRST, and no continuum tensor is subtracted. But it means the matched gauge sector has *no* validation yet, and the gauge-coupling and gauge-field normalization matching cannot start.

Precision budget. The report's cost extrapolation (about 0.12, 0.95, 6.35 s per loop point at $L = 8, 10, 12$; roughly 87 serial days for a naive 8×2^{12}, six-regulator, three-route, continuum-plus-lattice $L = 12$ run, from a single timing point) makes plain Python integrands the bottleneck for the highest-order contacts. The $L = 10, 12$ values scale steeply ($\sim 10^4$ and $\sim 10^6$) and are dominated by large cancellations at ~ 4 points per scramble, so their relative errors of 10–17% should not be treated as stable.

Scope risks. (a) Power-divergent pieces depend on a, so bare mass shifts are not cutoff independent; test matched amplitudes or observables under spacing refinement. (b) $g_0 = 0.65$ is a benchmark, not $g_0(a)$. (c) The IR regulator breaks BRST in the omitted gauge/ghost sectors, so the Higgs-only sectors are consistent only if the regulator effects vanish as $\lambda_{\mathrm{IR}} \to 0$, which is exactly what the extrapolation assumes.

11 [Added] Open-issues register

ID	Sector	Issue	Closure criterion
O1	Scalar contacts	4-pt nonexceptional vs. zero-momentum differ by $\sim 3\sigma$	agreement within combined error after higher-order projector or extra routes
O2	Gauge-linked 4-pt	held-out residuals $\sim 1.4\sigma$, same sign	stable intercepts across 3/4/5 windows plus extra regulators
O3	All gauge-linked	no model-error estimate for IR extrapolation	test alternative forms (with $x \log x$) and wider regulator range
O4	$L = 6$	only 4×64 points	production-level paired run
O5	$L = 8$	paired window shifts $\sim 2.4\sigma$; contact pilot	window and regulator-extension study; scalar-only pairing
O6	$L = 10, 12$	4 points per scramble; errors unstable	optimized integrand, declared production precision
O7	Gauge two-point	continuum tensor missing; IR mass breaks BRST	continuum tensor + same projectors; Ward closure after $\lambda_{\mathrm{IR}} \to 0$
O8	Mass coefficient	matched zero-momentum δm^2 absent (tadpole only)	gauge-linked 2-point at zero momentum with common regulator
O9	Gauge coupling	$g_0(a)$ matching not specified	separate gauge-sector calculation
O10	Backgrounds	only radial; no nonzero gauge background	needed for gauge-field matching
O11	Integration	vertex module not consumed by matching integrator/determinant	integrate or document as external validation
O12	Cost	~ 87 serial days naive $L = 12$	vectorize/parallelize before launch

Table 14: [Added] Open items, each tied to a measurable closure condition.

12 Corrected next steps (from the report)

1. **Production precision for $L = 8, 10, 12$.** Refine the fixed-route continuum and lattice integrations to a declared production precision with enough independent scrambles and points to stabilize both intercepts and errors. Keep paired lattice/continuum scrambles, the same six-regulator sequence,

compare 3-/4-/5-point fits and test sensitivity to extra regulator values before adopting any projection. Optimize or parallelize first (naive $L = 12$ run ~ 87 serial days; rough single-point extrapolation, not a benchmark).

2. **Continuum gauge two-point tensor.** Compute continuum pure-gauge, ghost and matter contributions with the same external momenta and P_T, P_L projectors; apply UV renormalization; compare to the raw-lattice loop plus Haar term.

3. **Common IR sequence and Ward/Slavnov–Taylor tests** [11, 3] on every massless line; test regulator cancellation and stability under fit-window and QMC refinement; separate QMC, regulator-model and UV-tail uncertainties.

4. **Finite differences only after paired stability.** Do not demand cutoff-independent bare mass shifts; test matched amplitudes or physical observables under spacing refinement.

5. **Write the coefficient JSON only when the selected scope is complete,** with keys `delta_ZH`, `delta_m2`, `delta_lambda`, `delta_gamma`, `delta_delta`, `delta_c10`, `delta_c12`; run the basis evaluator on independent smooth backgrounds; keep `coefficients=null` and the full-matching flag false until the scope passes its checks.

Full gauge-field and gauge-coupling matching additionally requires nonzero background gauge fields and operator projections.

13 [Added] Suggested roadmap with gates

The workstreams below organize the report's steps by dependency. Gates are suggested acceptance conditions, not results.

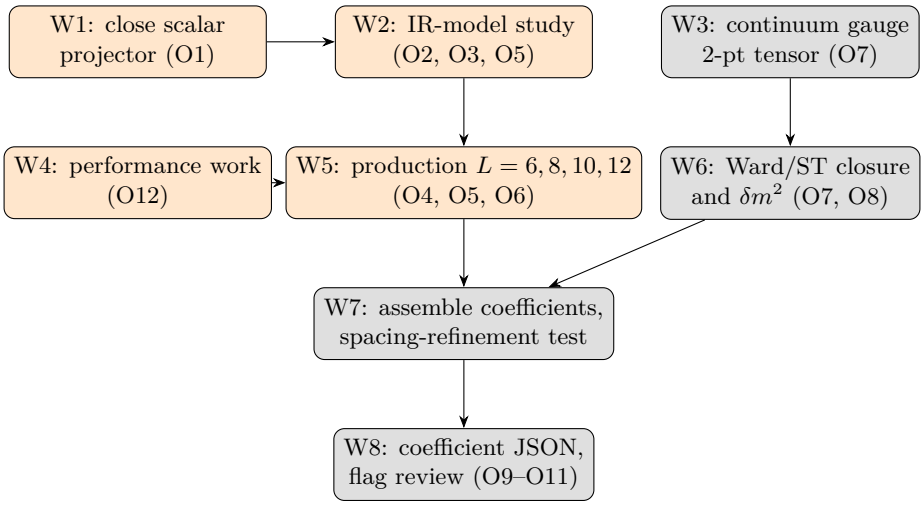


Figure 23: [Added] Dependency graph of the remaining work. Orange: can start now (partial results exist); gray: blocked on upstream items.

Stream	Starts from	Deliverable	Suggested gate
W1	scalar 4-pt contact vs. zero-momentum (O1)	repeat with more routes and a higher-order projector	agreement within 2σ on both 4- and 6-pt
W2	4-pt held-out residuals; $L = 8$ shifts	compare polynomial vs. polynomial-plus-$x \log x$ fits on a wider regulator range	model-selected intercepts agree within QMC error
W3	Wilson/ghost vertices (audited)	continuum gauge/ghost/matter tensor with P_T, P_L and $\overline{\text{MS}}$ subtraction	transversality at the matter-check level
W4	timing of 0.12–6.35 s per point	vectorized/parallel integrand	agreement with current code at roundoff; stated speed-up
W5	W1, W2, W4	paired runs at $L = 6, 8, 10, 12$ at declared precision	relative QMC SE target set in advance; window shifts within 1σ
W6	W3	Ward/ST closure as $\lambda_{\text{IR}} \to 0$; matched δm^2	normalized residual consistent with zero within errors
W7	W5, W6	coefficient assembly; test under spacing refinement	matched amplitudes stable in a
W8	W7	coefficient JSON with exact key set	independent smooth-background evaluator passes; flag decision documented

Table 15: [Added] Workstreams with suggested gates. Target precisions are left to the project to declare before launching W5.

14 [Added] Index of diagrams

Sector	Figures	Status
Status map and finite-matching pipeline	Secs. 1	overview
Conventions	legend, propagator table	—
Plaquette, scalar link	Sec. 4	definition
Scalar-gauge, Hessian, Wilson, ghost, potential, counterterm vertices	Sec. 5	audited
Routings and projectors	Sec. 5	audited
Higgs two-point, scalar tadpole, scalar rings	Secs. 6–8	complete for scope
Trace-log ring, 4-Higgs, 6–12-Higgs classes	Secs. 6, 9	pilots
Gauge two-point and Ward check	Secs. 6, 9	matter check passes; rest open
Roadmap graph	Sec. 13	planning

Table 16: [Added] Where each diagram family appears in this merged edition.

Part IV
Reference material

15 Artifact index

Solver and derivations.
`Higgs_fluctuation_solver_stabilized.py` (`make_su2_link_operators`); `SU2_Physical_Prefactor_Progress_Content.tex`; `Higgs_SU2_Gauge_Coupled_Hessian_Report_20260930.tex`; `SU2_Matching_Coefficient_Progress_Report_20261006.tex`.

Vertex module and audit.
`su2_lattice_vertices.py`; `su2_nonlinear_vertex_audit.py`; `su2_nonlinear_vertex_tables_20261007_conserving.json` (the `.npz` is referenced upstream but not present in the workspace).

Routing deck.
`su2_full_gauged_projection_setup.py`, `su2_full_gauged_projection_audit.py`, `su2_full_gauged_projection_tables_20261008.json`, `su2_full_gauged_projection_audit_20261008.json` (the `.npz` is not present; the Oct 18 in-memory deck was not written).

One-loop evaluators.
`su2_one_loop_step2_partial_20261008.py` and `su2_one_loop_step2_partial_20261008_corrected.json`.

Four-Higgs gauge sector.
`su2_four_point_gauge_sector_20261008.py`; `su2_four_point_gauge_sector_20261008_ircheck.json`; `su2_four_point_gauge_sector_20261011_ircheck.json`; `su2_four_point_gauge_sector_20261011_extended_ircheck.json`; `su2_four_point_fit_window_audit_20261012.py` and `su2_four_point_fit_window_audit_20261012.json`.

Oct 18 higher-point and gauge two-point.
`su2_higher_point_gauge_insertions_20261018.py`; `su2_gauge_two_point_followup_20261018.py`; `su2_continuum_uv_asymptotic_paired_20261018.json`; `su2_higher_point_gauge_insertions_20261018_L6_pilot.json`, `_L6_msbar_qmc.json`, `_L8_msbar_qmc_pilot.json`, `_L8_msbar_qmc_power4.json`; `su2_gauge_two_point_followup_20261018_pilot.json`, `_qmc_followup_haar.json`, `_qmc_power7_haar.json`.

Oct 19.
`su2_higher_point_gauge_insertions_20261019_L8_msbar_qmc_power5.json`; the `L10` and `L12` `power2` pilot and `8rep_followup` files.

Oct 23–24.
`outputs_archive.zip` containing `home/merritt/su2_paired_integration_driver.py`, `home/merritt/run_results/run_manifest.json`, `home/merritt/run_results/results.json`; run ID `20261023T173421Z`.

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